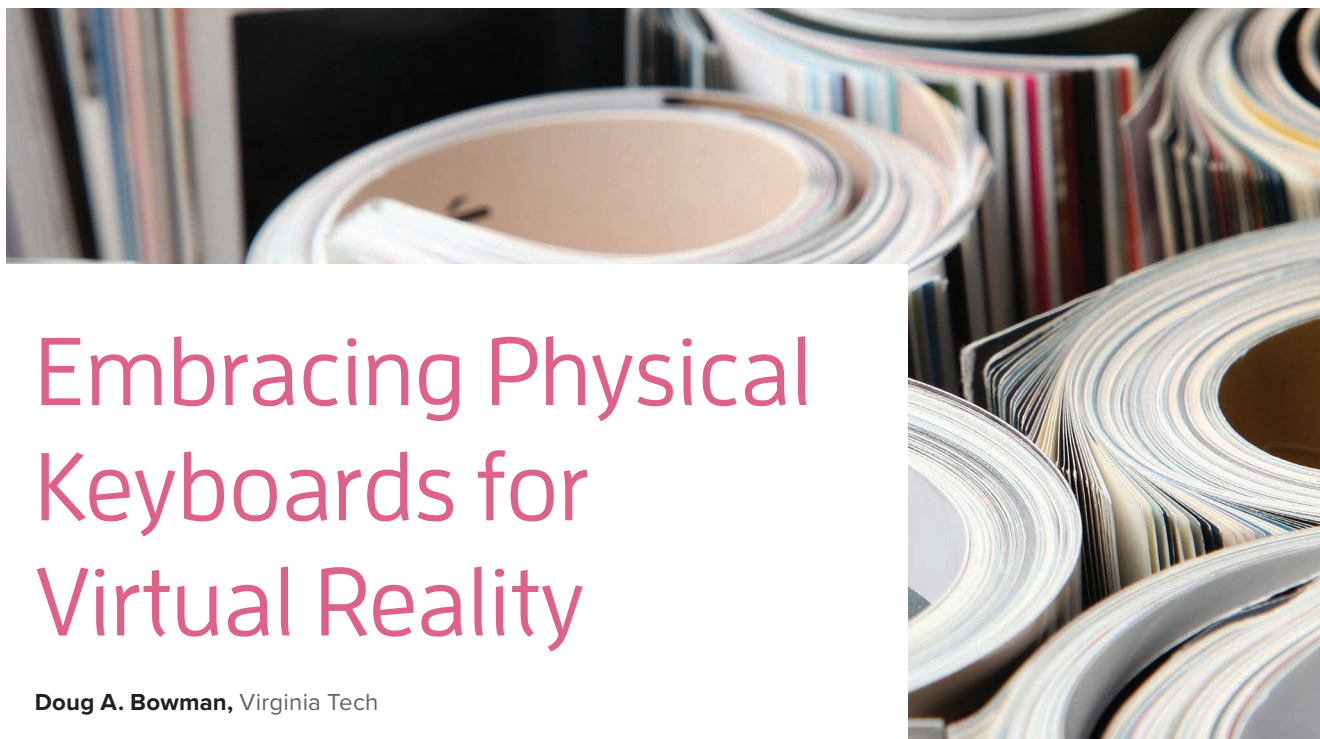




Embracing Physical Keyboards for Virtual Reality

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This installment of Computer's series highlighting the work published in IEEE Computer Society journals comes from IEEE Transactions on Visualization and Computer Graphics.

As virtual reality (VR) systems become more popular and affordable, users are rediscovering an old problem that remains challenging despite decades of research—how to enter text into VR applications. While text entry might be unnecessary in VR games and entertainment, there is a surprising number of use cases, such as entering a file name when saving a 3D scene, writing a note to a collaborator who will enter the virtual world later, or searching for a particular string in a document visualization. Moreover, as VR headsets improve, more people are considering doing office work in VR, where the number and size of

screens are unlimited. But virtual keyboards in VR are slow, gestures for text entry are hard to learn, and dictation systems are not appropriate for all usage contexts. By far, the best-performing technique is to use a physical keyboard like the one attached to your PC, and recent work has demonstrated that, with accurate finger and keyboard tracking,

typing on a physical keyboard in VR can result in very good performance. The problem, however, is that physical keyboards are a one-trick pony; they can be used only for entering text. Keyboards in VR seem like a necessary but undesirable peripheral.

In “ReconViguration: Reconfiguring Physical Keyboards in Virtual Reality,” Daniel Schneider and colleagues¹ show that, in VR, physical keyboards can become much more than a device for entering characters. Their article, which also won the best paper award at IEEE ISMAR 2019, proposes that physical keyboards can be adapted for use in a variety of VR applications and interactions capable of taking advantage of their unique affordances (for example, dozens of unique keys, haptic feedback for both touching and depressing the keys, and broad familiarity

to users). The key idea is that VR allows designers to magically change both the appearance and function of any physical object, including the keyboard.

(that is, pressing any key underneath the image). This is a fine example of using the whole keyboard for application-specific tasks when text entry is not needed.

that future, repurposing the keyboard for powerful interactions beyond text entry would be highly desirable. This work also reminds us that when it comes to VR/AR interfaces, explicitly breaking assumptions about how something is used can be an effective design strategy. 

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This enables a wide variety of interesting interfaces. For example, simply by shuffling the order in which letters are assigned to keys, a keyboard can be used to enter passwords or other private information into a VR application without fear of “shoulder surfing.” The authors show that while this slows down typing speed considerably, users see great utility in this technique. Another application discussed in the article uses the keyboard as if it were a grid of larger buttons, allowing direct access to particular images in a photo browser. The images are overlaid on the keyboard, and selecting a photo is as easy as “touching” it

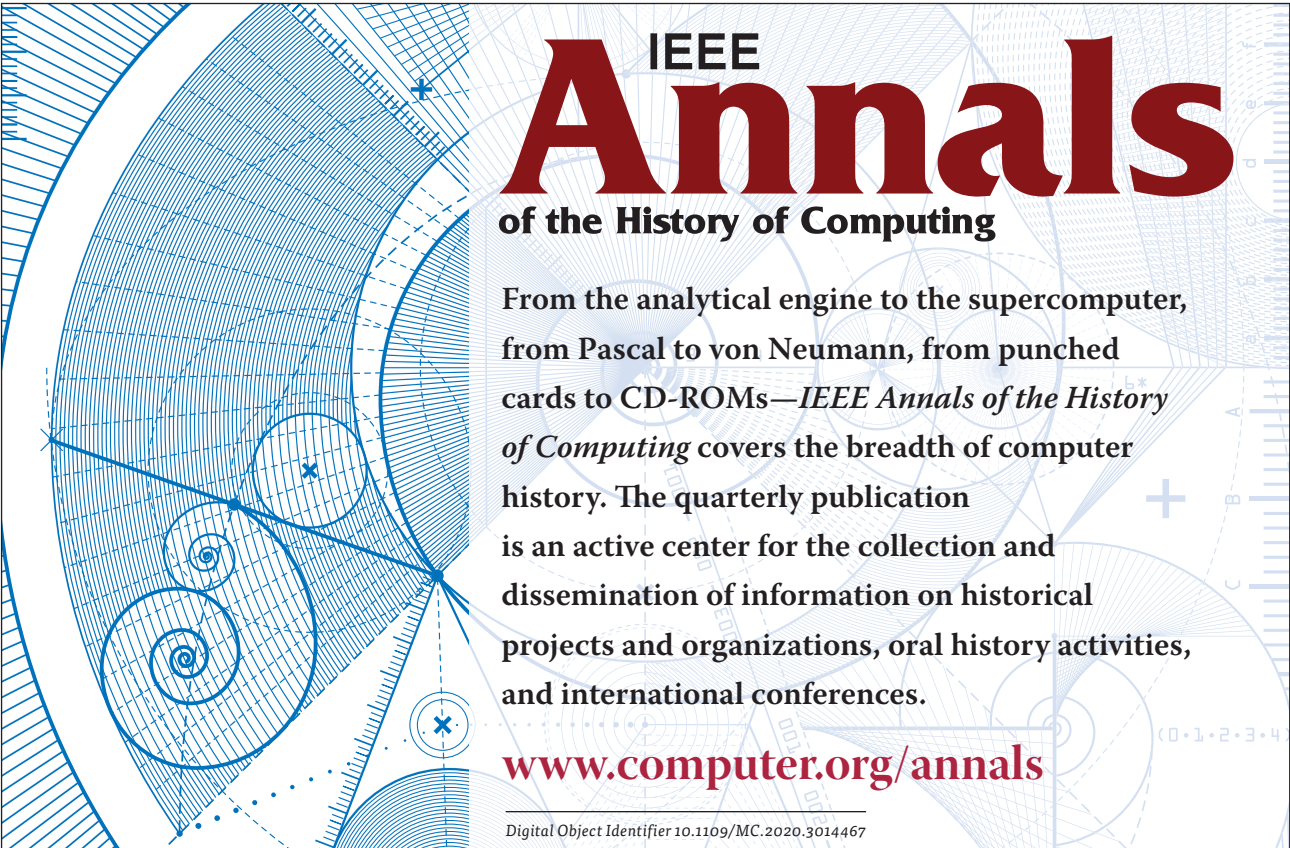
As a final example, the authors show how the keyboard can even be used for continuous input—a row of keys can become a slider, allowing the user to scrub through a video or go directly to a particular spot on the timeline.

The work of Schneider et al.¹ could also be applied to augmented reality (AR), which might become an even more ubiquitous technology for productivity work. One can certainly imagine future workers replacing their laptops with AR glasses and a lightweight, wireless keyboard, allowing them to work from anywhere without sacrificing screen real estate. In

REFERENCE

1. D. Schneider et al., “ReconViguation: Reconfiguring physical keyboards in virtual reality,” *IEEE Trans. Vis. Comput. Graphics*, vol. 25, no. 11, pp. 3190–3201, 2019. doi: 10.1109/TVCG.2019.2932239.

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